Chapter #8: Differential and Multistage Amplifiers

from **Microelectronic Circuits** Text

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Introduction

- **IN THIS CHAPTER YOU WILL LEARN:**
  - The essence of the operation of the MOS and the bipolar differential amplifiers: how they reject common-mode noise or interference and amplify differential signals.
  - The analysis and design of MOS and BJT differential amplifiers.
  - Differential amplifier circuits of varying complexity; utilizing passive resistive loads, current-source loads, and cascodes - the building blocks studied in Chapter 7.
  - An ingenious and highly popular differential-amplifier circuit that utilizes a current-mirror load.

Introduction

- The differential-pair of differential-amplifier configuration is widely used in IC circuit design.
  - One example is input stage of op-amp.
  - Another example is emitter-coupled logic (ECL).
- Technology was invented in 1940's for use in vacuum tubes – the basic differential-amplifier configuration was later implemented with discrete bipolar transistors.
- However, the configuration became most useful with invention of modern transistor / MOS technologies.

8.1. **The MOS Differential Pair**

- Figure 8.1: MOS differential-pair configuration.
- Two matched transistors (Q_1 and Q_2) joined and biased by a constant current source I.
- FET's should not enter triode region of operation.

8.1. **The MOS Differential Pair**

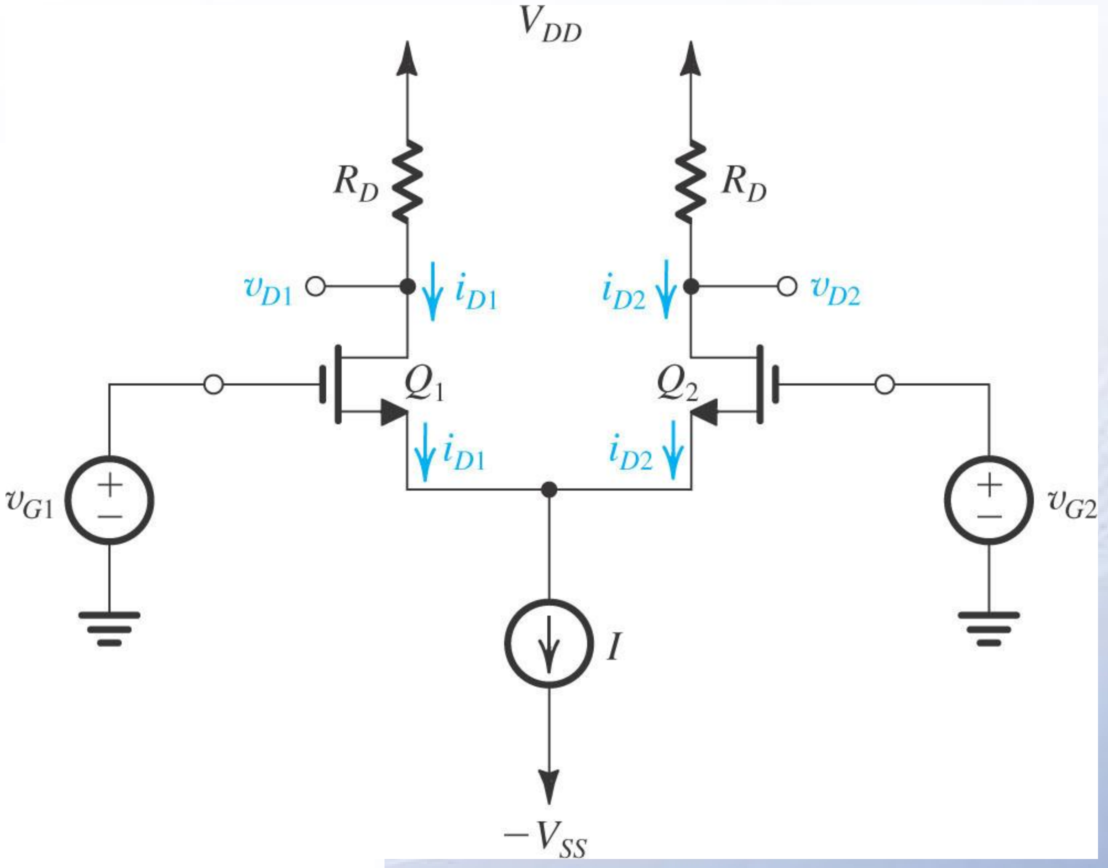
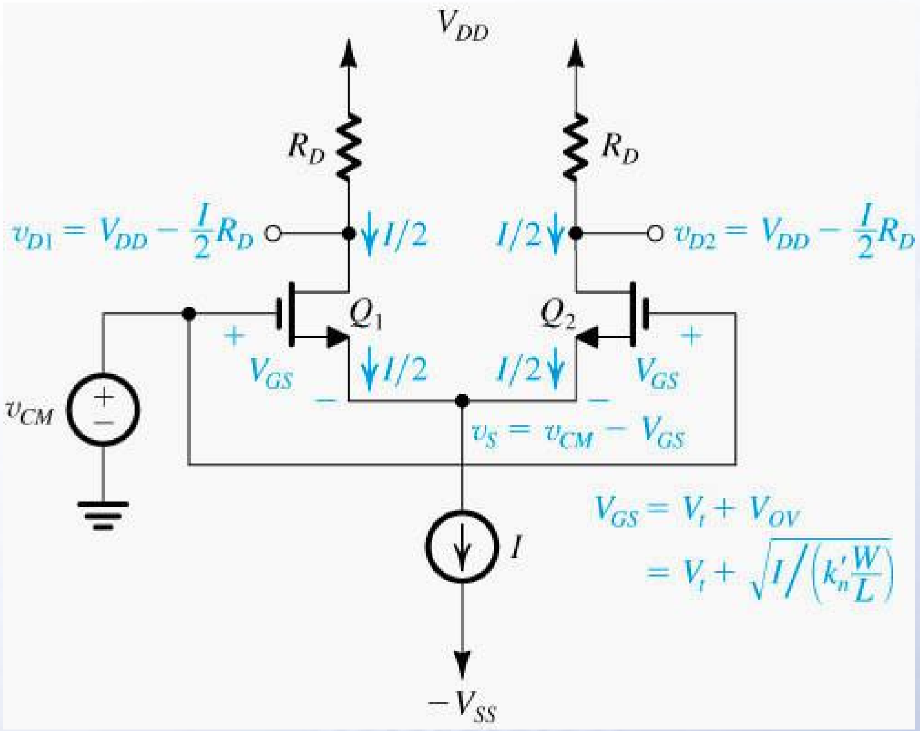


Figure 8.1: The basic MOS differential-pair configuration.

8.1.1. **Operation with a Common-Mode Input Voltage**



- Consider case when two gate terminals are joined together.
  - Connected to a **common-mode voltage** (V_{CM}).
  - $v_{G1} = v_{G2} = V_{CM}$
- Q_1 and Q_2 are matched.
- Current I will divide equally between the two transistors.
  - $I_{D1} = I_{D2} = I/2$, $V_S = V_{CM} - V_{GS}$
    - where V_{GS} is the **gate-to-source voltage**.

8.1.1. **Operation with a Common-Mode Input Voltage**

- Equations (8.2) through (8.8) describe this system, if channel-length modulation is neglected.
  - Note specification of input common-mode range (V_{CM}).

(8.2) $$\frac{I}{2} = \frac{1}{2} k_n' \frac{W}{L} (V_{GS} - V_t)^2$$

(8.3) $$V_{OV} = V_{GS} - V_t$$

(8.4) $$\frac{I}{2} = \frac{1}{2} k_n' \frac{W}{L} V_{OV}^2$$

(8.5) $$V_{OV} = \sqrt{\frac{I}{k_n'} \frac{L}{W}}$$

(8.6) $$v_{D1} = v_{D2} = V_{DD} - \frac{I}{2} R_D$$

(8.7) $$\max(V_{CM}) = V_t + V_{DD} - \frac{I}{2} R_D$$

(8.8) $$\min(V_{CM}) = -V_{SS} + V_{CS} + V_t$$

V_{CS} is voltage needed across the current source

Example 8.1

For the MOS differential pair with a common-mode voltage V_{CM} applied, as shown in Fig. 8.2, let $V_{DD} = V_{SS} = 1.5$ V, $k_n'(W/L) = 4$ mA/V^2, $V_t = 0.5$ V, $I = 0.4$ mA, and $R_D = 2.5$ kΩ, and neglect channel-length modulation. Assume that the current source I requires a minimum voltage of 0.4 V to operate properly.

(a) Find V_{OV} and V_{GS} for each transistor.
(b) For $V_{CM} = 0$, find V_S, I_{D1}, I_{D2}, V_{D1}, and V_{D2}.
(c) Repeat (b) for $V_{CM} = +1$ V.
(d) Repeat (b) for $V_{CM} = -0.2$ V.
(e) What is the highest permitted value of V_{CM}?
(f) What is the lowest value allowed for V_{CM}?

Solution

(a) With $v_{G1} = v_{G2} = V_{CM}$, we see that $V_{GS1} = V_{GS2}$. Now, since the transistors are matched, I will divide equally between the two transistors,

$$I_{D1} = I_{D2} = \frac{I}{2}$$

Thus,

$$\frac{I}{2} = \frac{1}{2}k_n'(W/L)V_{OV}^2$$

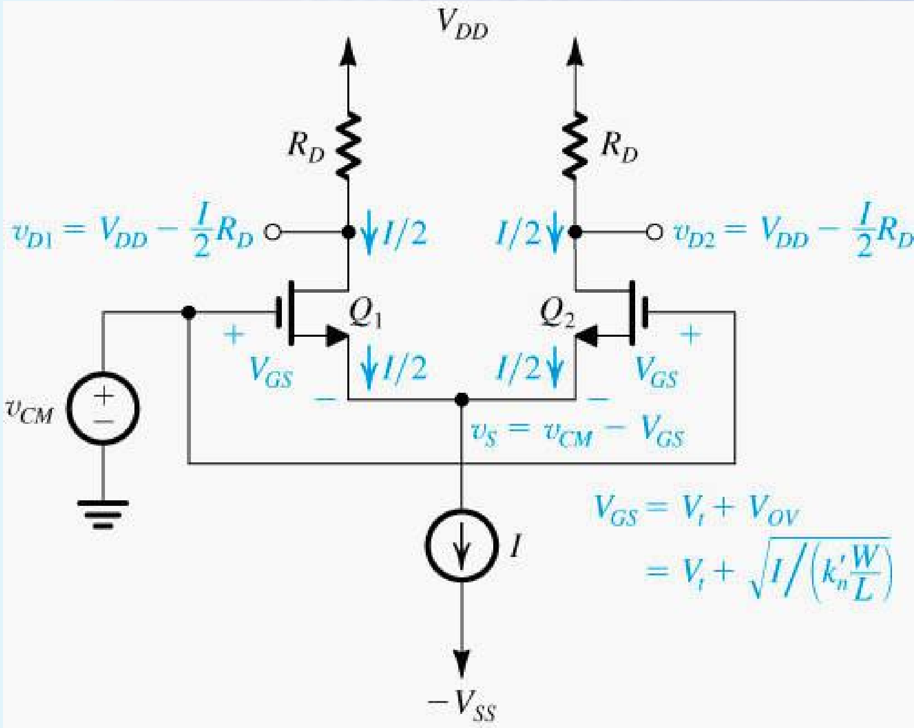
$$\frac{0.4}{2} = \frac{1}{2} \times 4V_{OV}^2$$

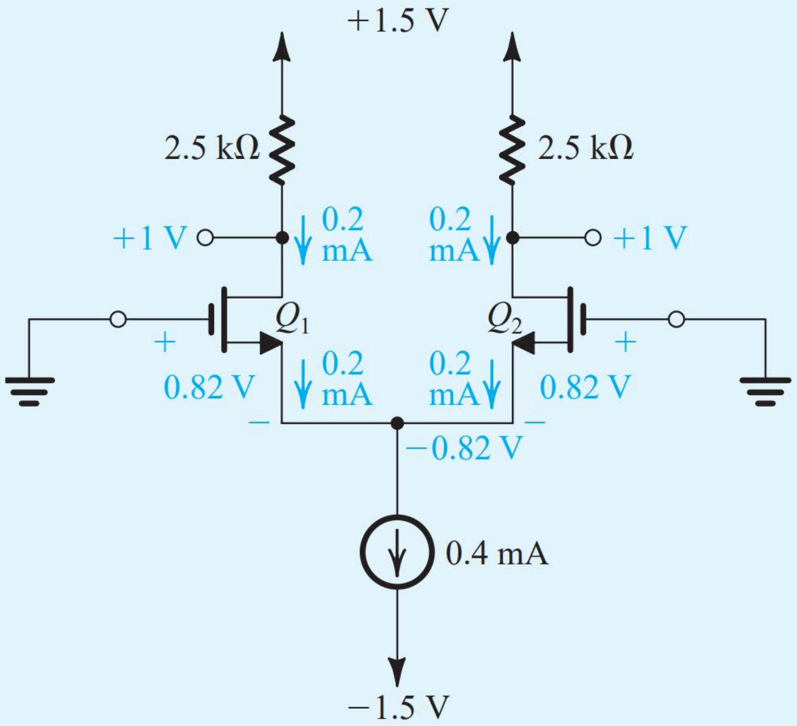
which results in

$$V_{OV} = 0.316 \text{ V}$$

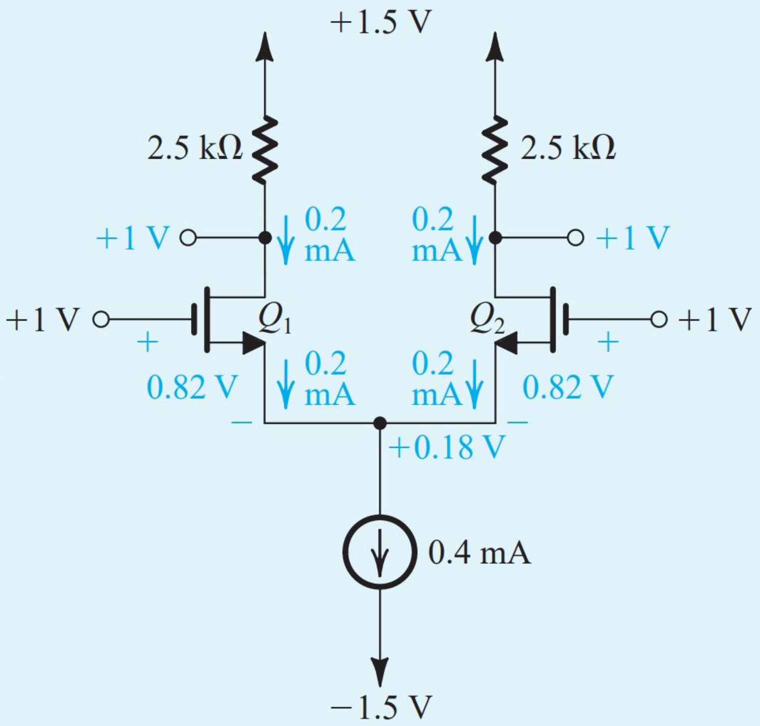
and thus,

$$V_{GS} = V_t + V_{OV} = 0.5 + 0.316 \simeq 0.82 \text{ V}$$

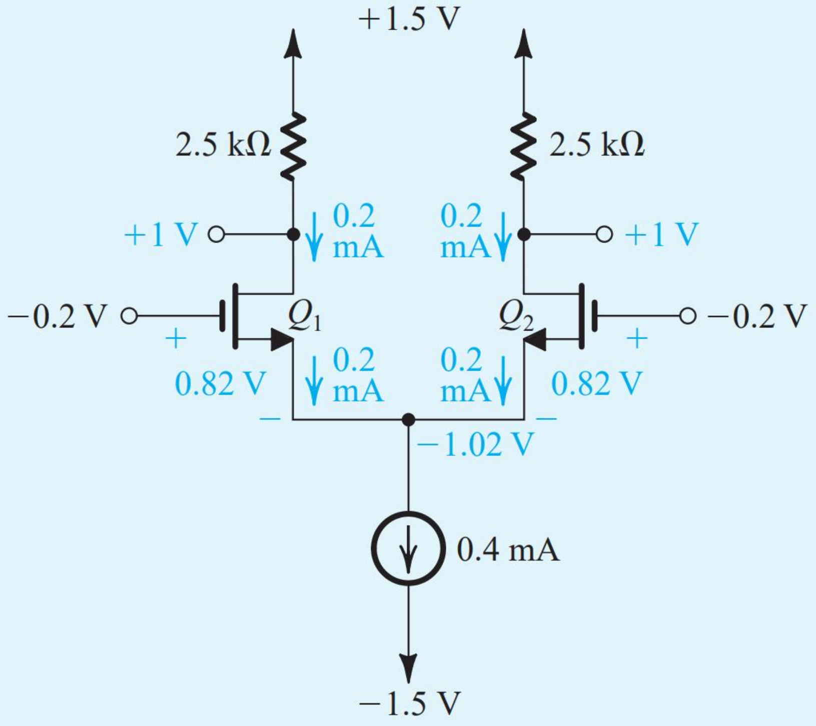



+1.5 V
2.5 kΩ
2.5 kΩ
+1 V
0.2 mA
0.2 mA
+1 V
Q1
Q2
+
0.82 V
−
0.2 mA
0.2 mA
+
0.82 V
−
−0.82 V
0.4 mA
−1.5 V

(a)


+1.5 V
2.5 kΩ
2.5 kΩ
+1 V
0.2 mA
0.2 mA
+1 V
+1 V
Q1
Q2
+1 V
+
0.82 V
−
0.2 mA
0.2 mA
+
0.82 V
−
+0.18 V
0.4 mA
−1.5 V

(b)


+1.5 V
2.5 kΩ
2.5 kΩ
+1 V
0.2 mA
0.2 mA
+1 V
−0.2 V
Q1
Q2
−0.2 V
+
0.82 V
−
0.2 mA
0.2 mA
+
0.82 V
−
−1.02 V
0.4 mA
−1.5 V

(c)

(b) The analysis for the case $V_{CM} = 0$ is shown in Fig. 8.3(a) from which we see that

$$V_S = V_G - V_{GS} = 0 - 0.82 = -0.82 \text{ V}$$

$$I_{D1} = I_{D2} = \frac{I}{2} = 0.2 \text{ mA}$$

$$V_{D1} = V_{D2} = V_{DD} - \frac{I}{2} R_D$$

$$= 1.5 - 0.2 \times 2.5 = 1 \text{ V}$$

(c) The analysis for the case $V_{CM} = +1$ V is shown in Fig. 8.3(b) from which we see that

$$V_S = V_G - V_{GS} = 1 - 0.82 = +0.18 \text{ V}$$

$$I_{D1} = I_{D2} = \frac{I}{2} = 0.2 \text{ mA}$$

$$V_{D1} = V_{D2} = V_{DD} - \frac{I}{2} R_D = 1.5 - 0.2 \times 2.5 = +1 \text{ V}$$

Observe that the transistors remain in the saturation region as assumed. Also observe that I_{D1}, I_{D2}, V_{D1}, and V_{D2} remain unchanged even though the common-mode voltage V_{CM} changed by 1 V.

(d) The analysis for the case $V_{CM} = -0.2$ V is shown in Fig. 8.3(c), from which we see that

$$V_S = V_G - V_{GS} = -0.2 - 0.82 = -1.02 \text{ V}$$

It follows that the current source I now has a voltage across it of

$$V_{CS} = -V_S - (-V_{SS}) = -1.02 + 1.5 = 0.48 \text{ V}$$

which is greater than the minimum required value of 0.4 V. Thus, the current source is still operating properly and delivering a constant current $I = 0.4$ mA and hence

$$I_{D1} = I_{D1} = \frac{I}{2} = 0.2 \text{ mA}$$

$$V_{D1} = V_{D2} = V_{DD} - \frac{I}{2} R_D = +1 \text{ V}$$

So, here again the differential circuit is not responsive to the change in the common-mode voltage V_{CM}.

(e) The highest value of V_{CM} is that which causes Q_1 and Q_2 to leave saturation and enter the triode region. Thus,

$$V_{CM\max} = V_t + V_D$$

$$= 0.5 + 1 = +1.5 \text{ V}$$

(f) The lowest value allowed for V_{CM} is that which reduces the voltage across the current source I to the minimum required of $V_{CS} = 0.4$ V. Thus,

$$V_{CM\min} = -V_{SS} + V_{CS} + V_{GS}$$

$$= -1.5 + 0.4 + 0.82 = -0.28 \text{ V}$$

Thus, the input common-mode range is

$$-0.28 \text{ V} \le V_{CM} \le +1.5 \text{ V}$$

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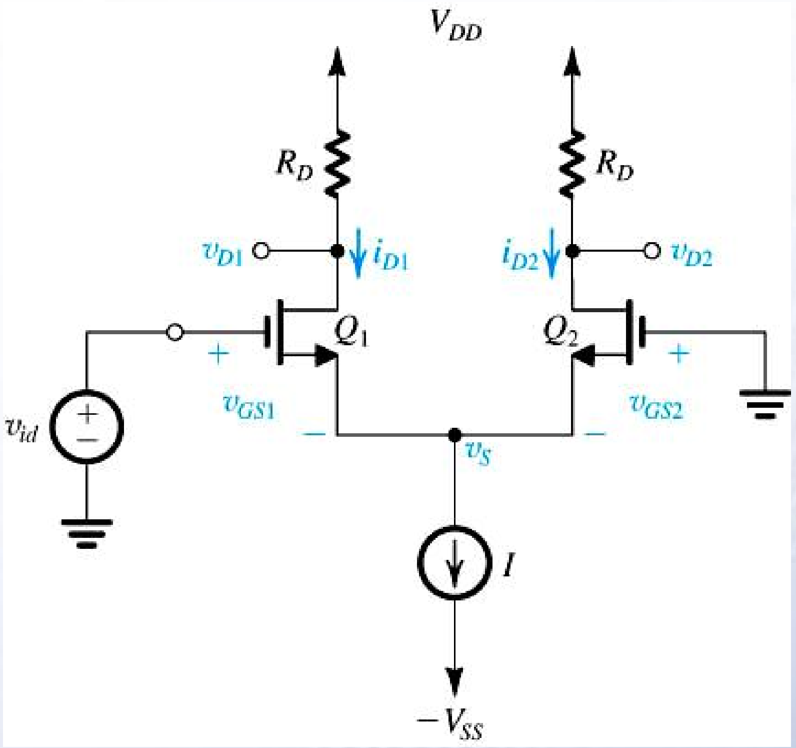
8.1.1. **Operation with a Common-Mode Input Voltage**

- Symmetry circuit.
- Common-mode voltage.
- Current *I* divides equally between two transistors.
- The difference between two drains is zero.
- *The differential pair rejects the common-mode input signals.*

8.1.2. **Operation with a Differential Input Voltage**



- If v_{id} applied to Q_1 and Q_2 is grounded, following conditions apply:
  - *If* $v_{id} = v_{GS1} - v_{GS2} > 0$ then $i_{D1} > i_{D2}$
  - *And* $v_o = v_{D2} - v_{D1} > 0$
- The opposite applies if Q_1 is grounded etc.
- The differential pair responds to a **difference-mode** or **differential input signals.**

8.1.2. **Operation with a Differential Input Voltage**

Q: What is maximum value of v_{id}

A: When $i_{D1} = I$, $v_S = -V_t$

$$(8.9)\quad I = \frac{1}{2}\left(k_n' \frac{W}{L}\right)\left(v_{GS1} - V_t\right)^2$$

$$(8.9)\quad v_{GS1} = V_t + \sqrt{\frac{2I}{k_n'(W/L)}}$$

$$(8.9)\quad v_{GS1} = V_t + \sqrt{2}V_{OV}$$

$$(8.10)\quad \max(v_{id}) = V_{GS1} + v_S$$

$$(8.10)\quad \max(v_{id}) = \sqrt{2}V_{OV}$$

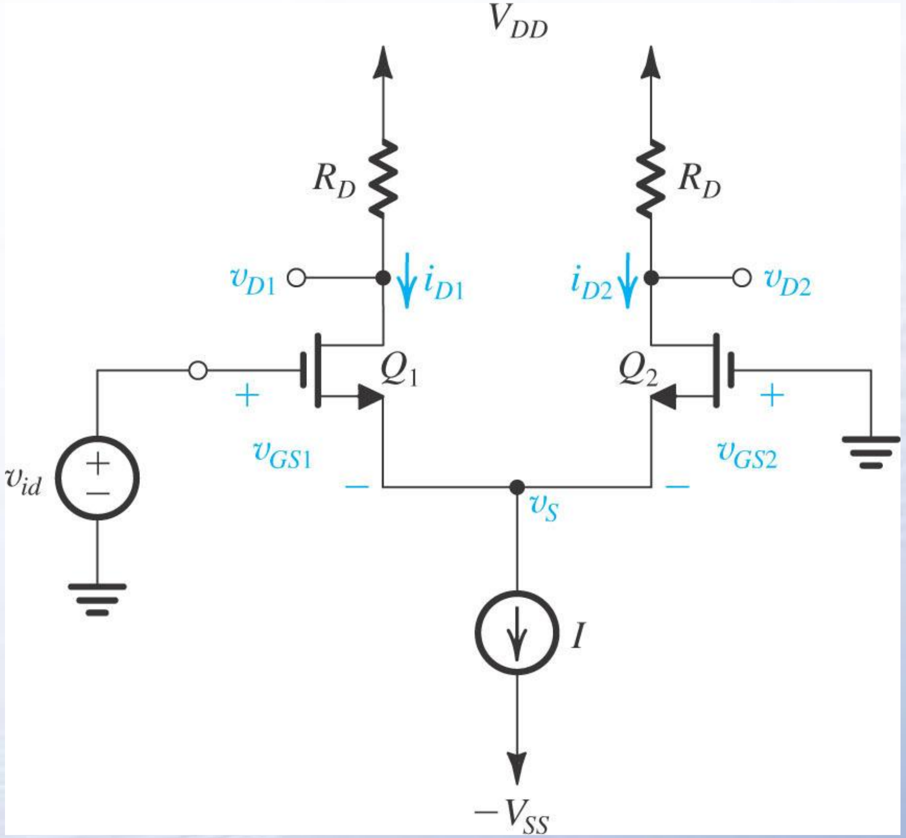


Figure 8.4: The MOS differential pair with a differential input signal v_{id} applied.

V_{OV} is the overdrive voltage corresponding to a drain current of $I/2$

8.1.2. **Operation with a Differential Input Voltage**

- Two input terminals connected to a suitable dc voltage V_{CM}.
- Bias current I of a "perfectly" symmetrical differential pair divides equally.
  - Zero voltage differential between the two drains.
- To steer the current completely to one side of the pair, a difference input voltage v_{id} of at most $2^{1/2}V_{OV}$ is needed.
- Range of differential-mode operation:

$$-\sqrt{2}V_{OV} \leq v_{id} \leq \sqrt{2}V_{OV}$$

8.1.3. **Large-Signal Operation**

- Objective is to derive expressions for drain current i_{D1} and i_{D2} in terms of differential signal $v_{id} = v_{G1} - v_{G2}$.
- Assumptions:
  - Perfectly Matched
  - Channel-length Modulation is Neglected
  - Load Independence
  - Saturation Region

8.1.3. **Large-Signal Operation**

- **step #1:** Express drain currents for Q_1 and Q_2.
- **step #2:** Take the square roots of both sides of both (8.11) and (8.12)
- **step #3:** Subtract (8.14) from (8.15) and perform appropriate substitution.
- **step #4:** Note the constant-current bias constraint.

$$(8.11)\quad i_{D1} = \frac{1}{2}k_n'\frac{W}{L}(v_{GS1} - V_t)^2$$

$$(8.12)\quad i_{D2} = \frac{1}{2}k_n'\frac{W}{L}(v_{GS2} - V_t)^2$$

$$(8.13)\quad \sqrt{i_{D1}} = \sqrt{\frac{1}{2}k_n'\frac{W}{L}}(v_{GS1} - V_t)$$

$$(8.14)\quad \sqrt{i_{D2}} = \sqrt{\frac{1}{2}k_n'\frac{W}{L}}(v_{GS2} - V_t)$$

$$(8.15)\quad v_{GS1} - v_{GS2} = v_{G1} - v_{G2} = v_{id}$$

8.1.3. **Large-Signal Operation**

- **step #5:** Simplify (8.15).
- **step #6:** Incorporate the constant-current bias.
- **step #7:** Solve (8.16) and (8.17) for the two unknowns – i_{D1} and i_{D2}.
  - Refer to (8.23) and (8.24).

(8.16) $\sqrt{i_{D1}} - \sqrt{i_{D2}} = \sqrt{\frac{1}{2}k_n' \frac{W}{L}} v_{id}$

(8.17) $i_{D1} + i_{D2} = I$

$$2\sqrt{i_{D1} i_{D2}} = I - \frac{1}{2}k_n' \frac{W}{L} v_{id}^2$$

(8.23) $i_{D1} = \frac{I}{2} + \left(\frac{I}{V_{OV}}\right)\left(\frac{v_{id}}{2}\right)\sqrt{1 - \left(\frac{v_{id}/2}{V_{OV}}\right)^2}$

(8.24) $i_{D2} = \frac{I}{2} - \left(\frac{I}{V_{OV}}\right)\left(\frac{v_{id}}{2}\right)\sqrt{1 - \left(\frac{v_{id}/2}{V_{OV}}\right)^2}$

8.1.3. **Large-Signal Operation**

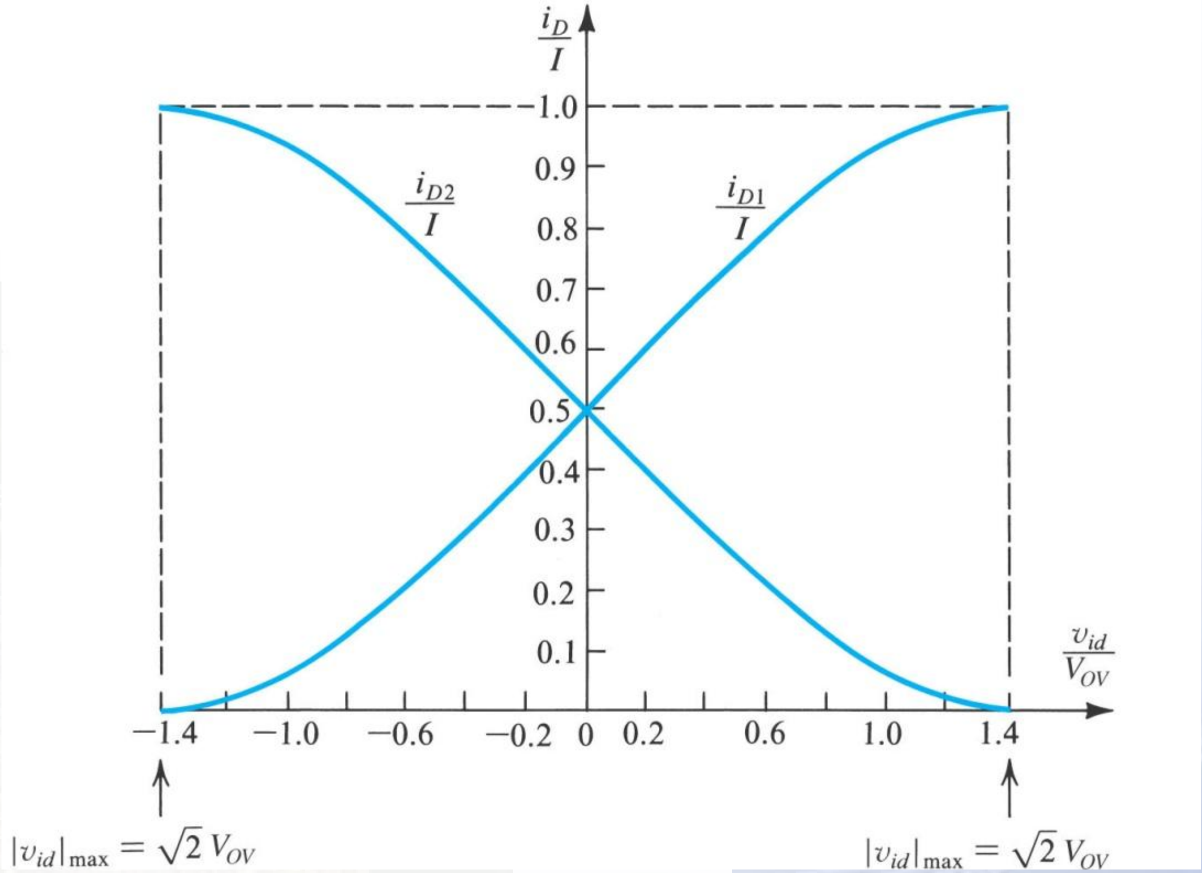


Figure 8.6: Normalized plots of the currents in a MOSFET differential pair. Note that V_{OV} is the overdrive voltage at which Q_1 and Q_2 operate when conducting drain currents equal to $I/2$, the equilibrium situation. Note that these graphs are universal and apply to any MOS differential pair

8.1.3. **Large-Signal Operation**

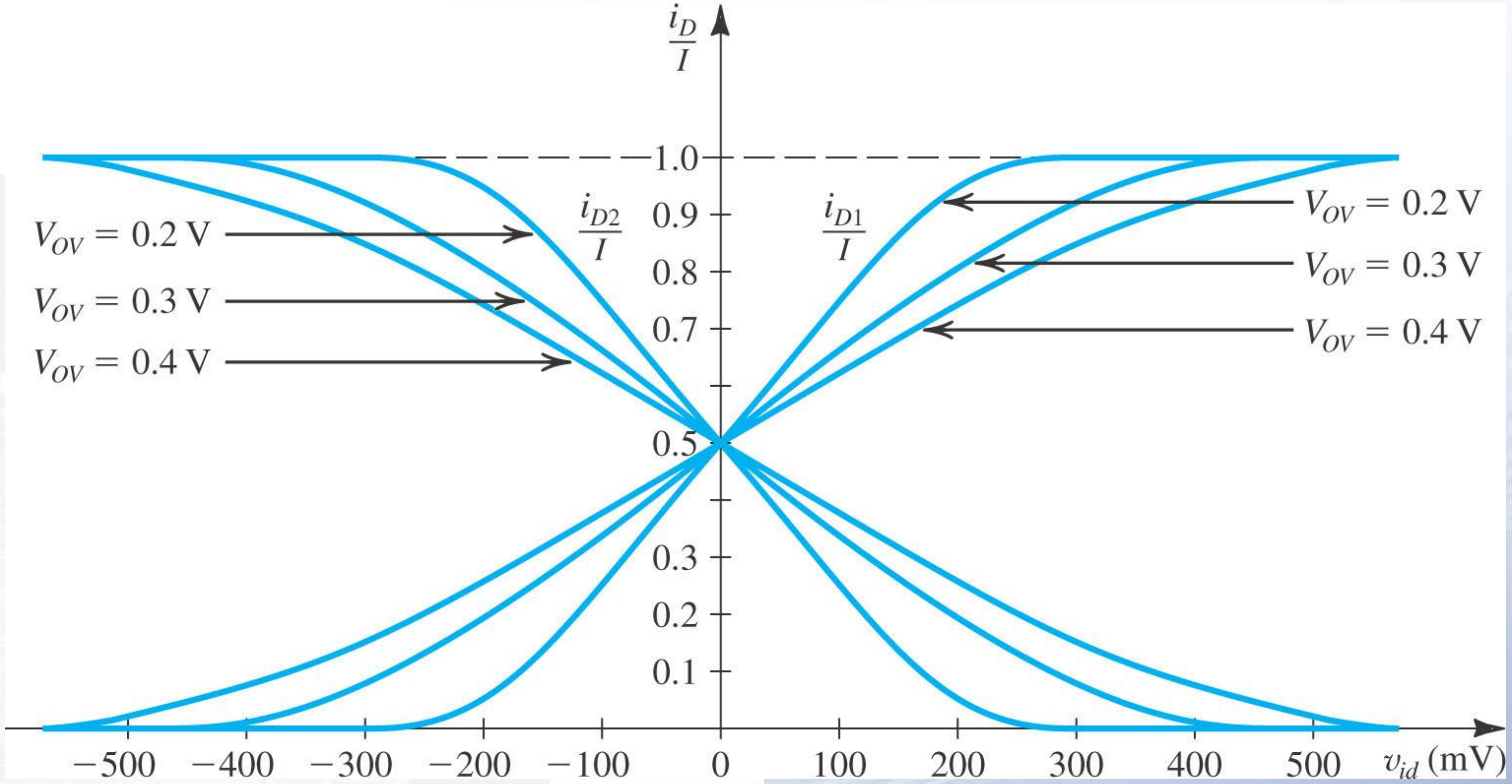


Figure 8.7: The linear range of operation of the MOS differential pair can be extended by operating the transistor at a higher value of V_{OV}.

8.1.3. **Large-Signal Operation**

$$(8.23)\ i_{D1} = \frac{I}{2} + \left(\frac{I}{V_{OV}}\right)\left(\frac{v_{id}}{2}\right)\sqrt{1-\left(\frac{v_{id}/2}{V_{OV}}\right)^2}$$

$$(8.24)\ i_{D2} = \frac{I}{2} - \left(\frac{I}{V_{OV}}\right)\left(\frac{v_{id}}{2}\right)\sqrt{1-\left(\frac{v_{id}/2}{V_{OV}}\right)^2}$$

$$\left(v_{id}/2\right) << V_{OV}$$

- Transfer characteristics of (8.23) and (8.24) are nonlinear.
- Linear amplification is desirable and v_{id} will be as small as possible.
- For a given value of V_{OV}, the only option is to keep $v_{id}/2$ much smaller than V_{OV}.

small-signal approximation

$$(8.25)\ i_{D1} \approx \frac{I}{2} + \left(\frac{I}{V_{OV}}\right)\frac{v_{id}}{2}$$

$$(8.26)\ i_{D2} \approx \frac{I}{2} - \left(\frac{I}{V_{OV}}\right)\frac{v_{id}}{2}$$

$$(8.27)\ i_d \approx \left(\frac{I}{V_{OV}}\right)\frac{v_{id}}{2}$$

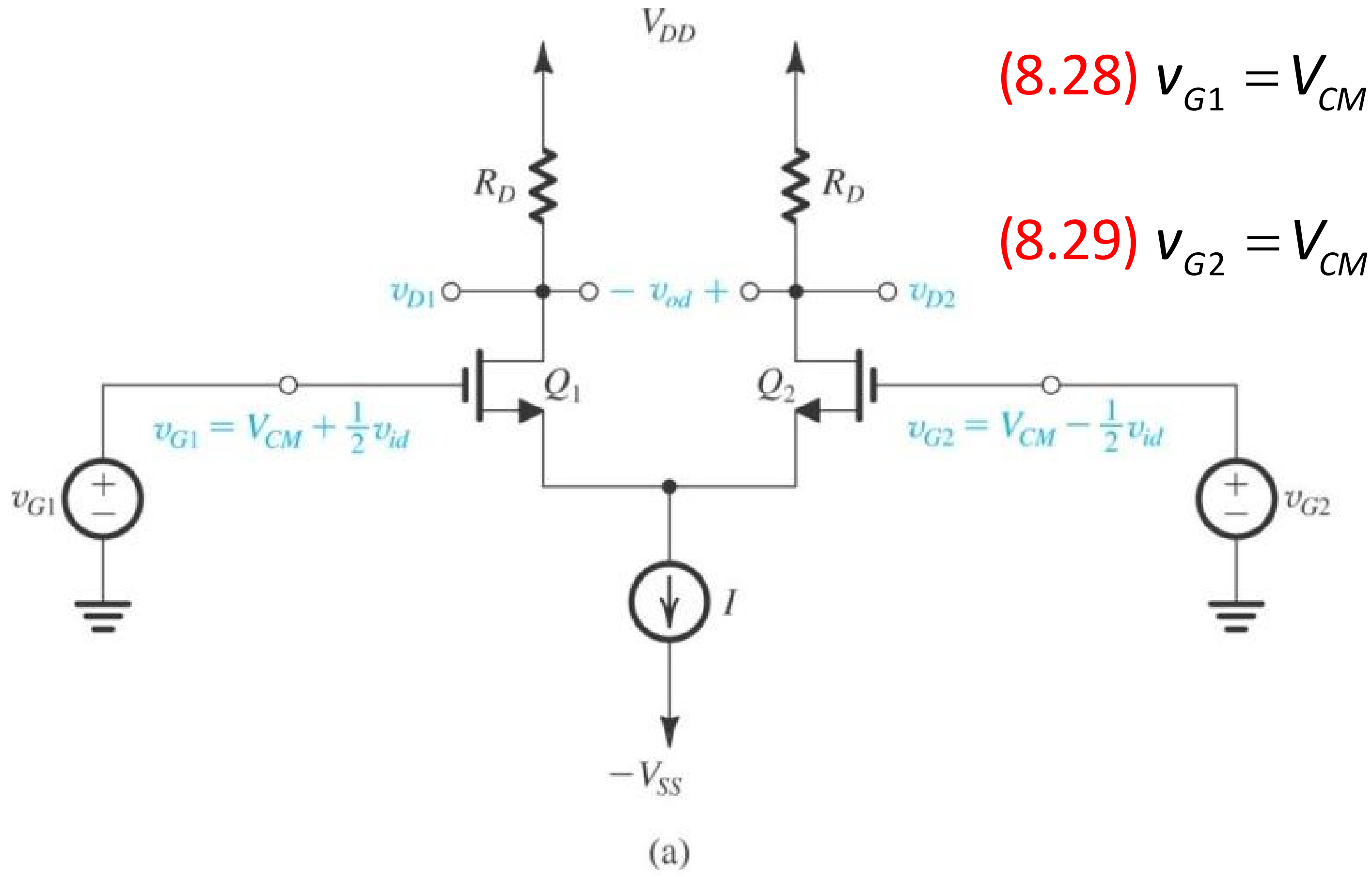
8.1.3. **Large-Signal Operation**

- Nonlinear curves.
- Maximum value of input differential voltage.
- When $v_{id} = 0$, two drain currents are equal to $I/2$.
- Linear segment.
- Linearity can be increased by increasing overdrive voltage.
- Price paid is a reduction in gain(current I is kept constant).

8.2. **Small-Signal Operation of the MOS Differential Pair**

- Linear amplifier
- Differential gain
- Common-mode gain
- Common-mode rejection ratio(CMRR)
- Mismatch on CMRR

8.2. **Small-Signal Operation of the MOS Differential Pair**



$$\text{(8.28)}\quad v_{G1} = V_{CM} + \frac{1}{2}v_{id}$$

$$\text{(8.29)}\quad v_{G2} = V_{CM} - \frac{1}{2}v_{id}$$

Figure 8.8 Small-signal analysis of the MOS differential amplifier.
(a) The circuit with a common-mode voltage applied to set the dc bias voltage at the gates and with v_{id} applied in a complementary (or balanced) manner.

8.2.1. **Differential Gain**

- $v_{i1} = V_{CM} + v_{id}/2$ and $v_{i2} = V_{CM} - v_{id}/2$ causes a virtual signal ground to appear on the common-source (common-emitter) connection
- Current in Q_1 increases by $g_m v_{id}/2$ and the current in Q_2 decreases by $g_m v_{id}/2$.
- Voltage signals of $g_m(R_D||r_o)v_{id}/2$ develop at the two drains.

8.2. **Small-Signal Operation of the MOS Differential Pair**

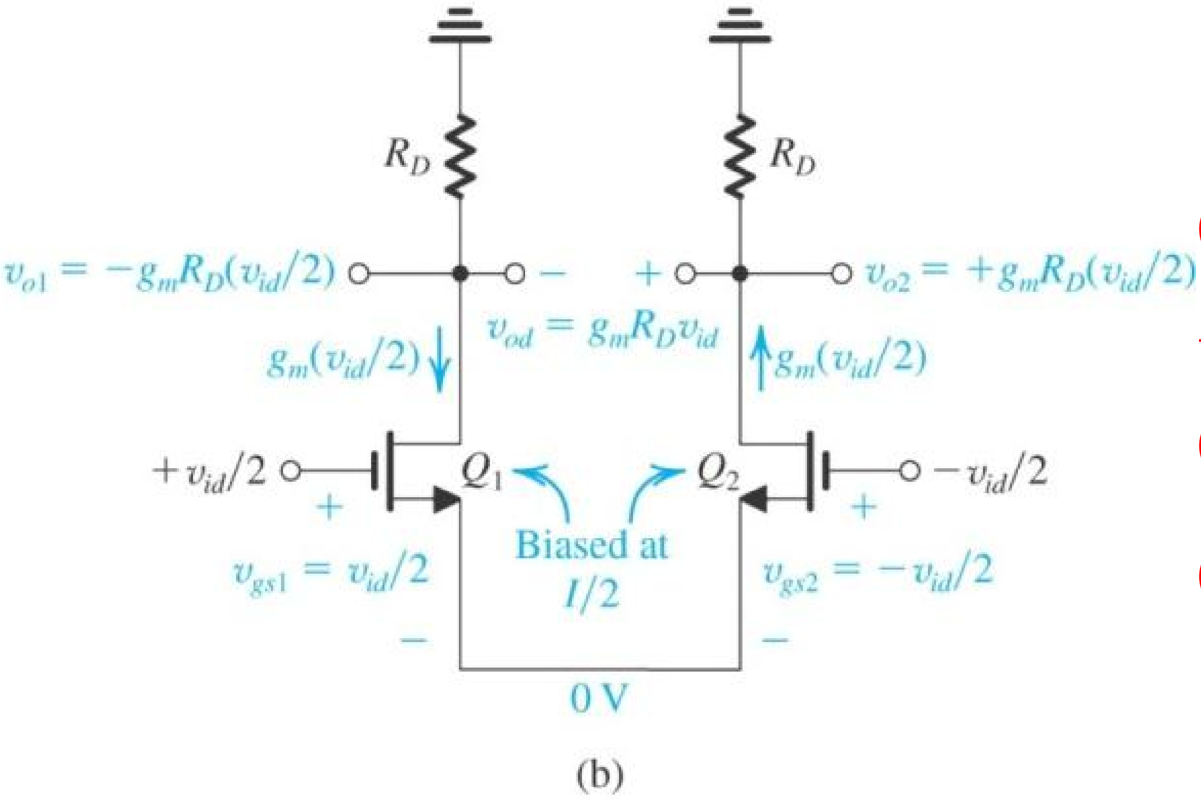


Figure 8.8 Small-signal analysis of the MOS differential amplifier.
(b) The circuit prepared for small-signal analysis.

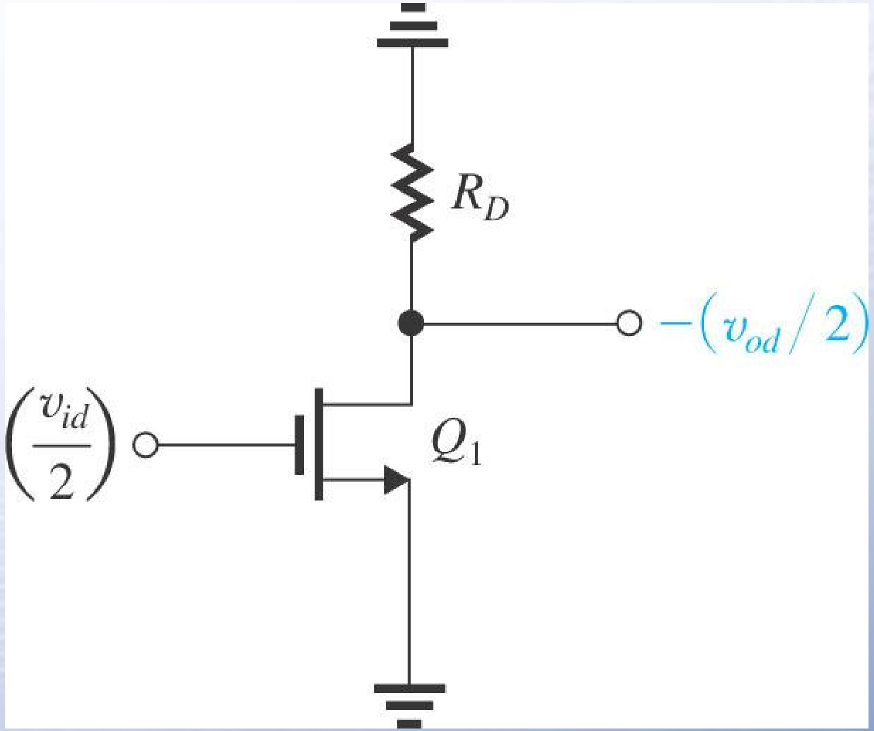
(8.30) $$g_m = \frac{2I_D}{V_{OV}} = \frac{2(I/2)}{V_{OV}} = \frac{I}{V_{OV}}$$

(8.31) $$v_{o1} = -g_m \frac{v_{id}}{2} R_D$$

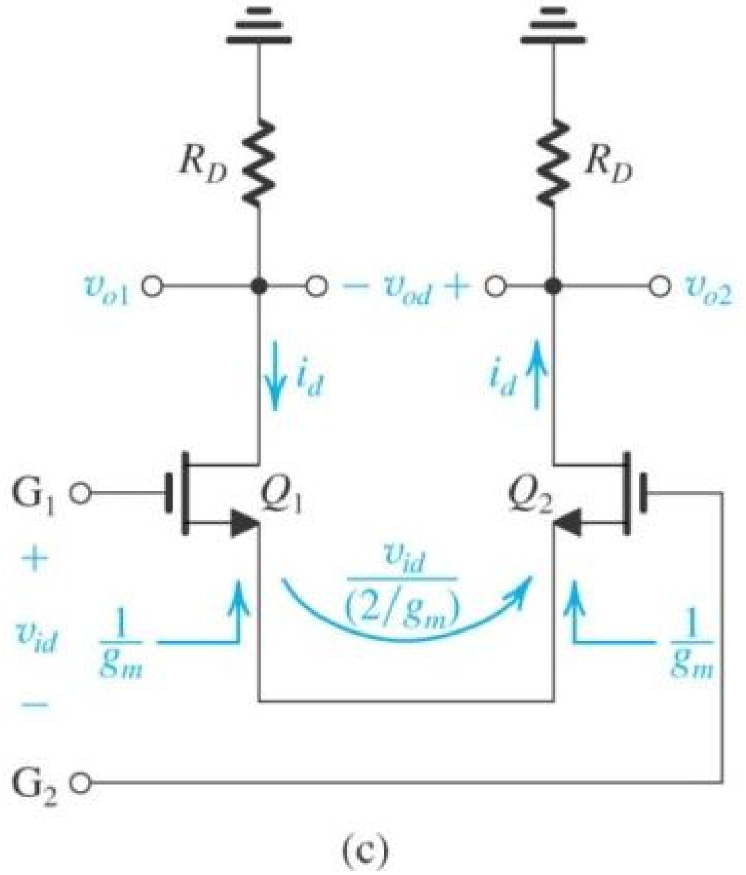
(8.32) $$v_{o2} = +g_m \frac{v_{id}}{2} R_D$$

8.2.2. **The Differential Half-Circuit**

- Figure 8.9 (right): The equivalent differential half-circuit of the differential amplifier of Figure 8.8.
- Here Q_1 is biased at $I/2$ and is operating at V_{OV}.
- This circuit may be used to determine the differential voltage gain of the differential amplifier $A_d = v_{od}/v_{id}$.



8.2. **Small-Signal Operation of the MOS Differential Pair**



(8.35) $A_d \equiv \frac{v_{od}}{v_{id}} = g_m R_D$

Figure 8.8 Small-signal analysis of the MOS differential amplifier.
(c) An alternative way of looking at the small signal operation of the circuit.

Example 8.2

Give the differential half-circuit of the differential amplifier shown in Fig. 8.10(a). Assume that Q_1 and Q_2 are perfectly matched. Neglecting r_o, determine the differential voltage gain $A_d \equiv v_{od}/v_{id}$.

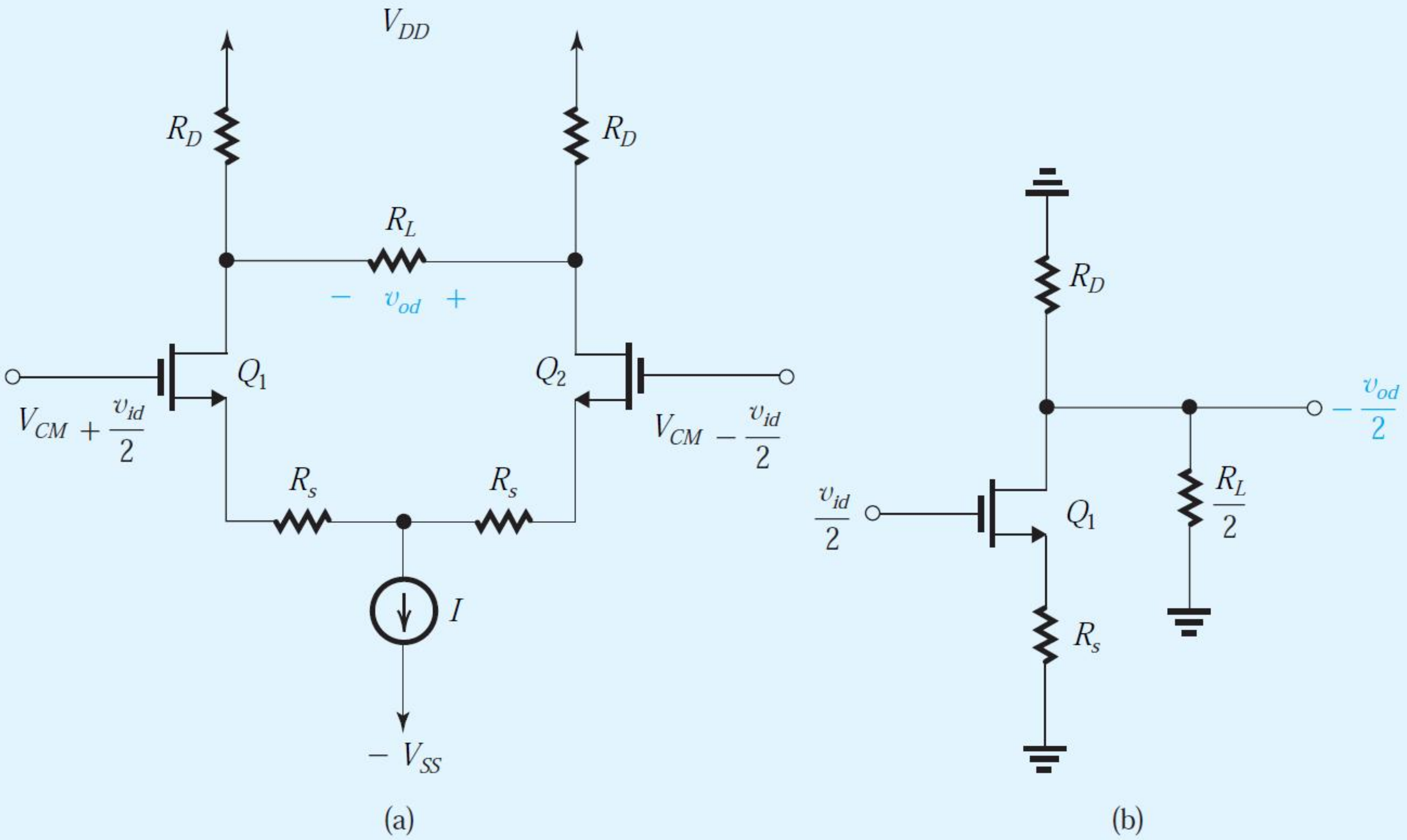


Figure 8.10 (**a**) Differential amplifier for Example 8.2. (**b**) Differential half-circuit.

Example 8.2

Solution

Since the circuit is symmetrical and is fed with v_{id} in a balanced manner, the differential half-circuit will be as shown in Fig. 8.10(b). Observe that because the line of symmetry passes through the middle of R_L, the half-circuit has a resistance $R_L/2$ connected between drain and ground. Also note that the virtual ground appears on the node between the two resistances R_s. As a result, the half-circuit has a source-degeneration resistance R_s.

Now, neglecting r_o of the half-circuit transistor Q_1, we can obtain the gain as the ratio of the total resistance in the drain to the total resistance in the source as

$$\frac{-v_{od}/2}{v_{id}/2} = -\frac{R_D \| (R_L/2)}{1/g_m + R_s}$$

with the result that

$$A_d \equiv \frac{v_{od}}{v_{id}} = \frac{R_D \| (R_L/2)}{1/g_m + R_s} \tag{8.37}$$

8.2.5. **Common-Mode Gain and Common-Mode Rejection ratio (CMRR)**

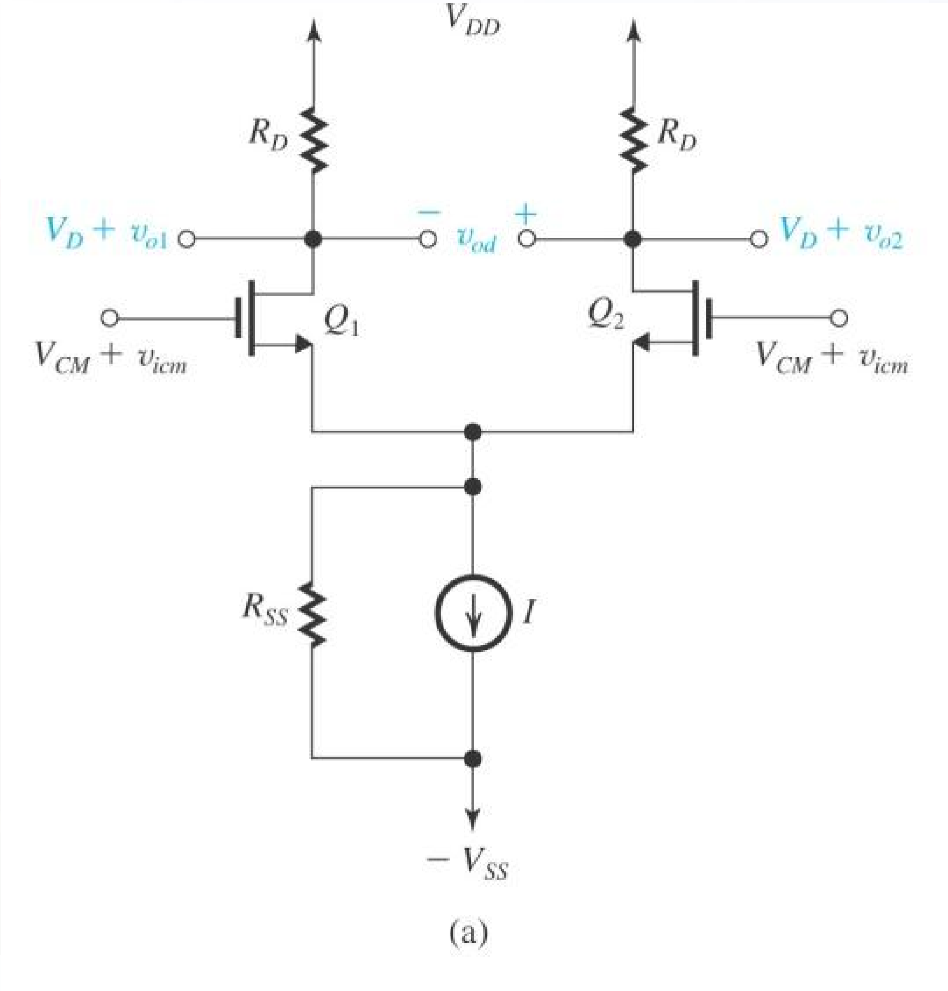
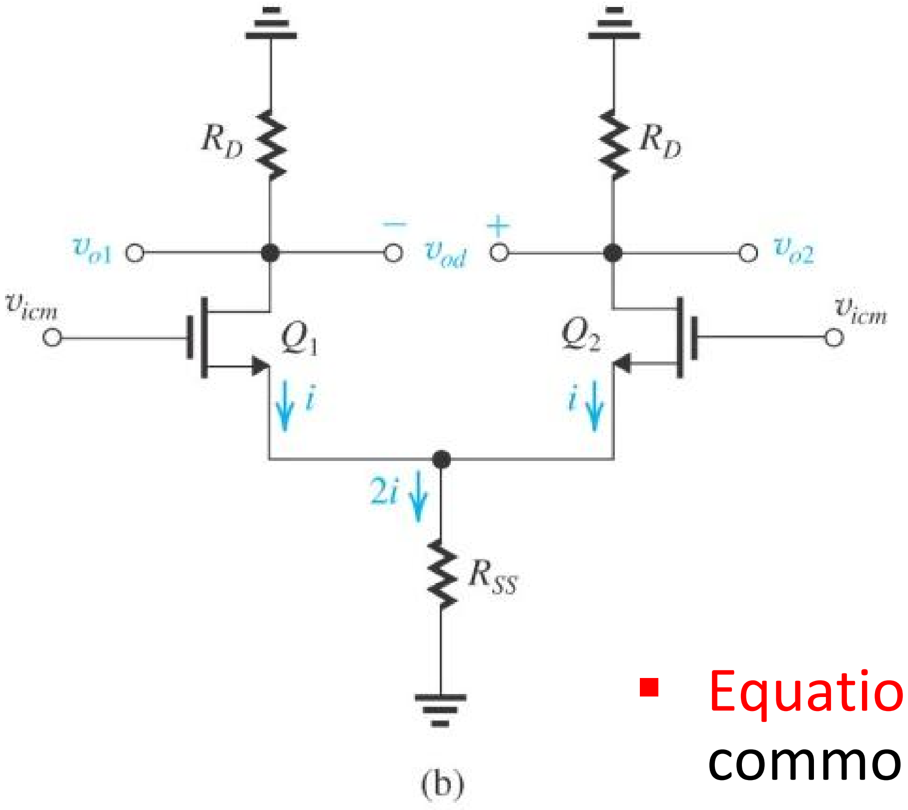


Figure 8.13 (a) A MOS differential amplifier with a common-mode input signal v_{icm} superimposed on the input dc common-mode voltage V_{CM}



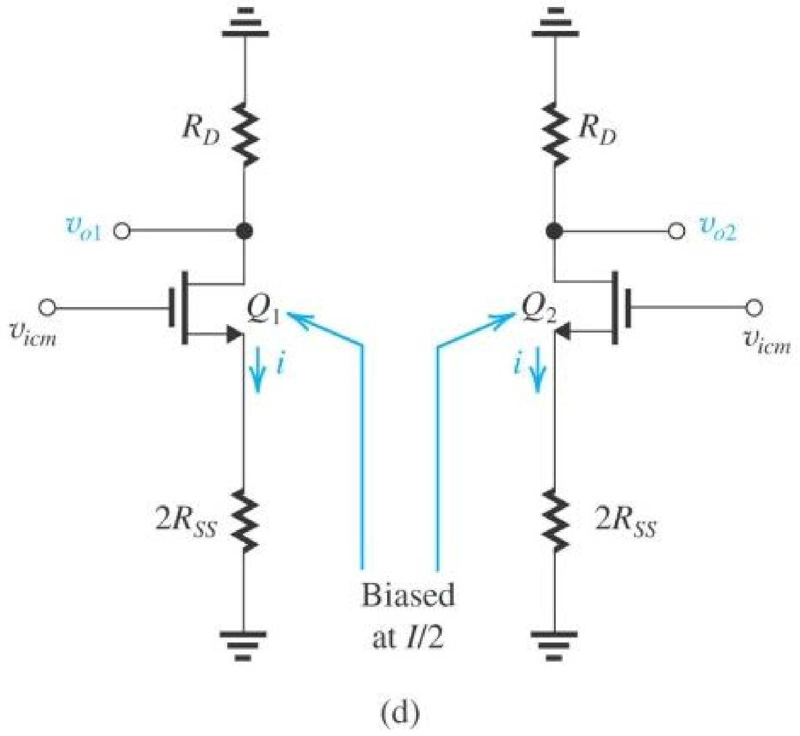
$$\text{(8.41)}\quad v_{icm} = \frac{i}{g_m} + 2iR_{SS}$$

$$\text{(8.42)}\quad i = \frac{v_{icm}}{1/g_m + 2R_{SS}}$$

$$\text{(8.43)}\quad v_{o1} \approx v_{o2} \approx -\frac{R_D}{1/g_m + 2R_{SS}} v_{icm}$$

- Equation (8.43) describes effect of common-mode signal (v_{icm}) on v_{o1} and v_{o2}.

Figure 8.13. **(b)** The amplifier circuit prepared for small-signal analysis.



(8.43) $$v_{o1} \approx v_{o2} \approx -\frac{R_D}{1/g_m + 2R_{SS}} v_{icm}$$

(8.44) $$v_{o1} \approx v_{o2} \approx -\frac{v_{icm} R_D}{2R_{SS}}$$

(8.45) $$v_{od} = v_{o2} - v_{o1} = 0$$

Figure 8.13 (d) The circuit in **(b)** split into its two halves; each half is called the "CM half circuit."

8.2.5. **Common-Mode Gain and Common-Mode Rejection ratio (CMRR)**

- When the output is taken single-ended, magnitude of common-mode gain is defined in (8.46).
- Taking the output differentially results in the perfectly matched case, in zero A_{cm} (infinite CMRR).

$$(8.46)\quad v_{o1} \approx -\frac{R_D}{2R_{SS}} v_{icm}$$

8.2.5. **Common-Mode Gain and Common-Mode Rejection ratio (CMRR)**

- Mismatches between the two sides of the pair make A_{cm} finite even when the output is taken differentially.
  - This is illustrated in (8.49).
- Corresponding expressions apply for the bipolar pair.
- Effect of R_D mismatch on CMRR is illustrated in (8.50)

$$(8.47)\quad v_{o2} = -\frac{\overbrace{R_D + \Delta R_D}^{RD\text{'s are mismatched}}}{2R_{SS}} v_{icm}$$

$$(8.48)\quad v_{od} = v_{o2} - v_{o1} = \frac{-\Delta R_D}{2R_{SS}} v_{icm}$$

$$(8.49)\quad A_{cm} \equiv \frac{v_{od}}{v_{icm}} = \frac{-\Delta R_D}{2R_{SS}} = \left(\frac{-R_D}{2R_{SS}}\right)\left(\frac{\Delta R_D}{R_D}\right)$$

$$(8.50)\quad CMRR \equiv \frac{|A_d|}{|A_{cm}|} = (2g_m R_{ss}) \Big/ \left(\frac{\Delta R_D}{R_D}\right)$$

Mismatch on CMRR

- Effect of g_m mismatch on CMRR

$$\frac{i_{d1}}{i_{d2}} = \frac{g_{m1}}{g_{m2}}$$

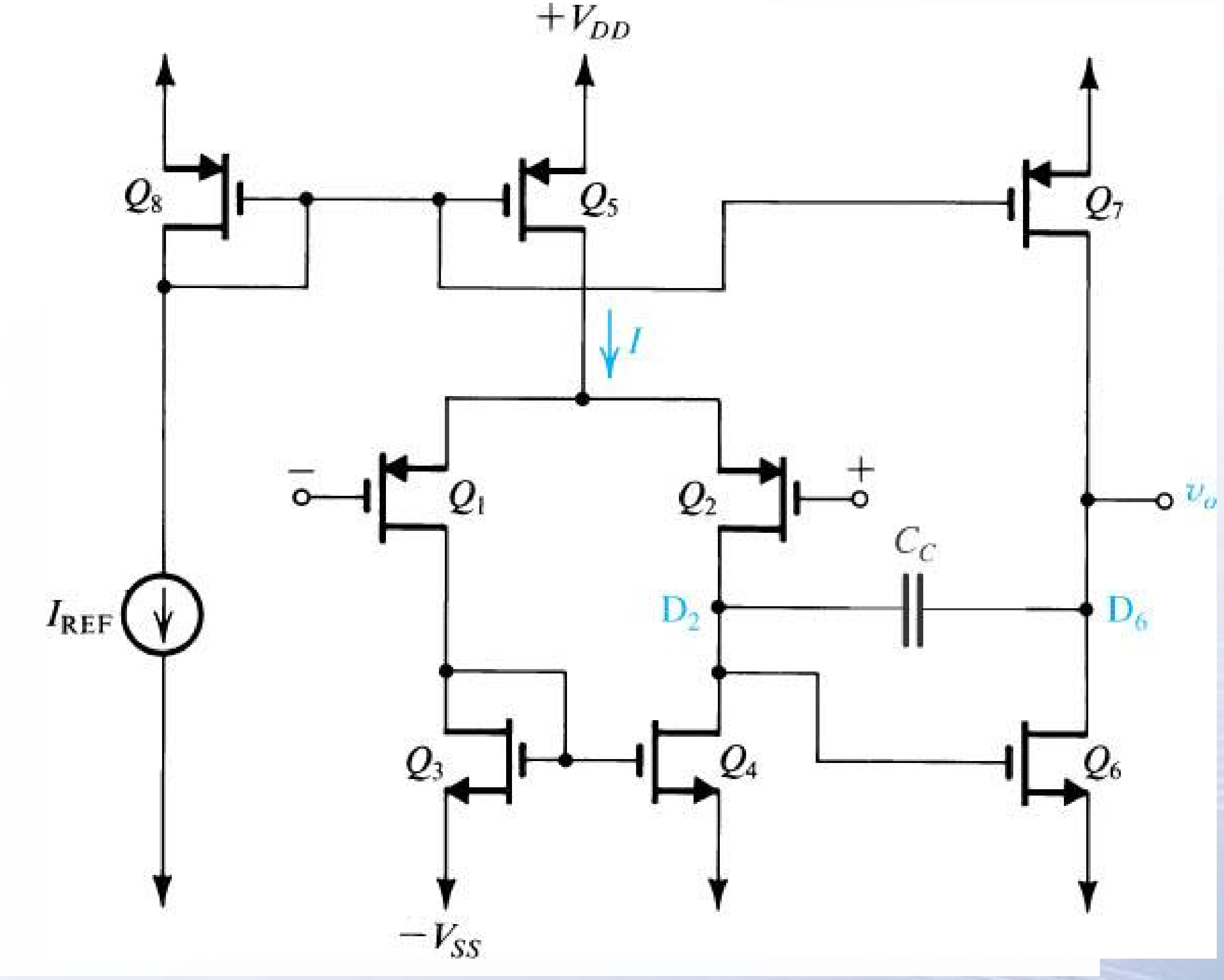
$$v_s \cong v_{icm}$$

$$i_{d1} - i_{d2} = \frac{\Delta g_m v_{icm}}{2 g_m R_{ss}}$$

$$CMRR = (2 g_m R_{ss}) \Big/ \left(\frac{\Delta g_m}{g_m}\right)$$

- A finite common-mode gain also can be made by a mismatch in the W/L values of Q1 and Q2.

Two-Stage CMOS Op-Amp Configuration



$$A_1 = -g_{m1}(r_{o2} \parallel r_{o4})$$

$$A_2 = -g_{m6}(r_{o6} \parallel r_{o7})$$

Figure 8.41 Two-stage CMOS op-amp configuration.

Summary

- The differential-pair or differential-amplifier configuration is most widely used building block in analog IC designs. The input stage of every op-amp is a differential amplifier.
- There are two reasons for preferring differential to single-ended amplifiers: 1) differential amplifiers are insensitive to interference and 2) they do not need bypass and coupling capacitors.
- For a MOS pair biased by a current source I, each device operates at a drain current of $I/2$ and a corresponding overdrive voltage V_{OV}. Each device has $g_m=1/V_{OV}$.

Summary

- With the two input terminals connected to a suitable dc voltage V_{CM}, the bias current I of a perfectly symmetrical differential pair divides equally between the two transistors of the pair, resulting in zero voltage difference between the two drains. To steer the current completely to one side of the pair, a difference input voltage v_{id} of at least $2^{1/2}V_{OV}$ is needed.
- Superimposing a differential input signal v_{id} on the dc common-mode input voltage V_{CM} such that $v_{I1} = V_{CM} + v_{id}/2$ and $v_{I2} = V_{CM} - v_{id}/2$ causes a virtual signal ground to appear on the common-source connection.

Summary

- The analysis of a differential amplifier to determine differential gain, differential input resistance, frequency response of differential gain, and so on is facilitated by employing the differential half-circuit which is a common-source transistor biased at $I/2$.
- An input common-mode signal vicm gives rise to drain (collector) voltage signals that are ideally equal and given by $-v_{icm}(R_D/2R_{SS})$, where R_{SS} is the output resistance of the current source that supplies the bias current I.

8.1 For an NMOS differential pair with a common-mode voltage V_{CM} applied, as shown in Fig. 8.2, let $V_{DD} = V_{SS} = 1.0$ V, $k'_n = 0.4$ mA/V^2, $(W/L)_{1,2} = 12.5$, $V_{tn} = 0.5$ V, $I = 0.2$ mA, $R_D = 10$ kΩ, and neglect channel-length modulation.

(a) Find V_{OV} and V_{GS} for each transistor.
(b) For $V_{CM} = 0$, find V_S, I_{D1}, I_{D2}, V_{D1}, and V_{D2}.
(c) Repeat (b) for $V_{CM} = +0.3$ V.
(d) Repeat (b) for $V_{CM} = -0.1$ V.
(e) What is the highest value of V_{CM} for which Q_1 and Q_2 remain in saturation?
(f) If current source I requires a minimum voltage of 0.2 V to operate properly, what is the lowest value allowed for V_S and hence for V_{CM}?

8.3 For the differential amplifier specified in Problem 8.1 let $v_{G2} = 0$ and $v_{G1} = v_{id}$. Find the value of v_{id} that corresponds to each of the following situations:

(a) $i_{D1} = i_{D2} = 0.1$ mA; (b) $i_{D1} = 0.15$ mA and $i_{D2} = 0.05$ mA; (c) $i_{D1} = 0.2$ mA and $i_{D2} = 0$ (Q_2 just cuts off); (d) $i_{D1} = 0.05$ mA and $i_{D2} = 0.15$ mA; (e) $i_{D1} = 0$ mA (Q_1 just cuts off) and $i_{D2} = 0.2$ mA. For each case, find v_S, v_{D1}, v_{D2}, and $(v_{D2} - v_{D1})$.

8.2 For the PMOS differential amplifier shown in Fig. P8.2 let $V_{tp} = -0.8$ V and $k'_p W/L = 4$ mA/V^2. Neglect channel-length modulation.

(a) For $v_{G1} = v_{G2} = 0$ V, find V_{OV} and V_{GS} for each of Q_1 and Q_2. Also find V_S, V_{D1}, and V_{D2}.
(b) If the current source requires a minimum voltage of 0.5 V, find the input common-mode range.

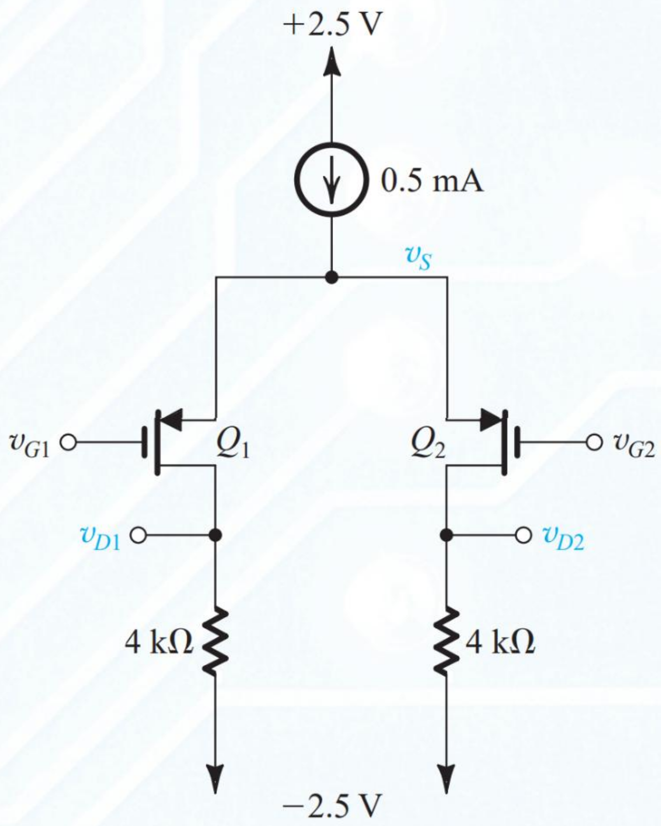


Figure P8.2

D 8.6 Design the circuit in Fig. P8.6 to obtain a dc voltage of +0.2V at each of the drains of Q_1 and Q_2 when $v_{G1} = v_{G2} = 0$ V. Operate all transistors at $V_{OV} = 0.2$ V and assume that for the process technology in which the circuit is fabricated, $V_{tn} = 0.5$ V and $\mu_n C_{ox} = 250\ \mu A/V^2$. Neglect channel-length modulation. Determine the values of R, R_D, and the W/L ratios of Q_1, Q_2, Q_3, and Q_4. What is the input common-mode voltage range for your design?

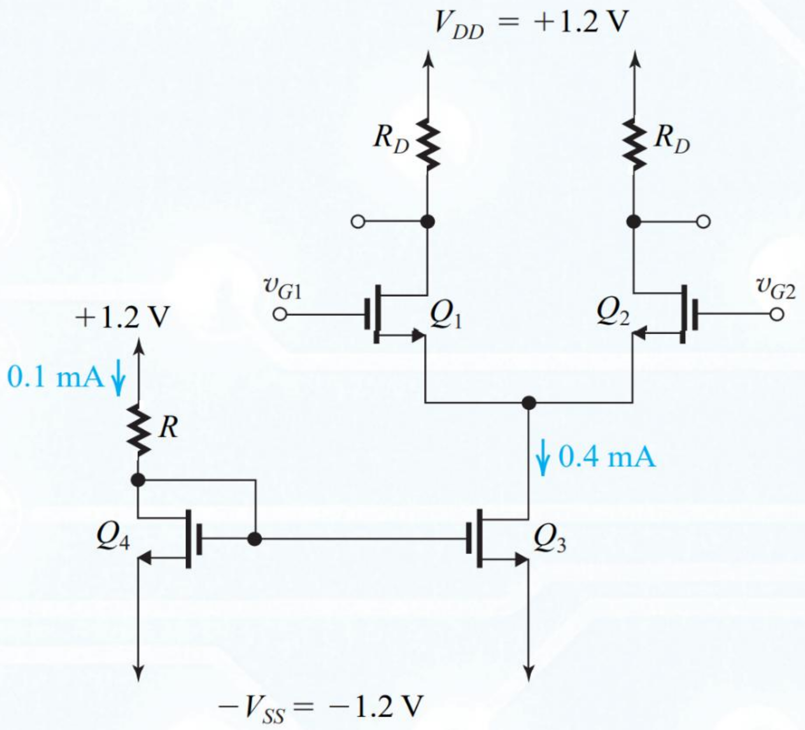


Figure P8.6

8.21 Find the differential half-circuit for the differential amplifier shown in Fig. P8.21 and use it to derive an expression for the differential gain $A_d \equiv v_{od}/v_{id}$ in terms of g_m, R_D, and R_s. Neglect the Early effect. What is the gain with $R_s = 0$? What is the value of R_s (in terms of $1/g_m$) that reduces the gain to half this value?

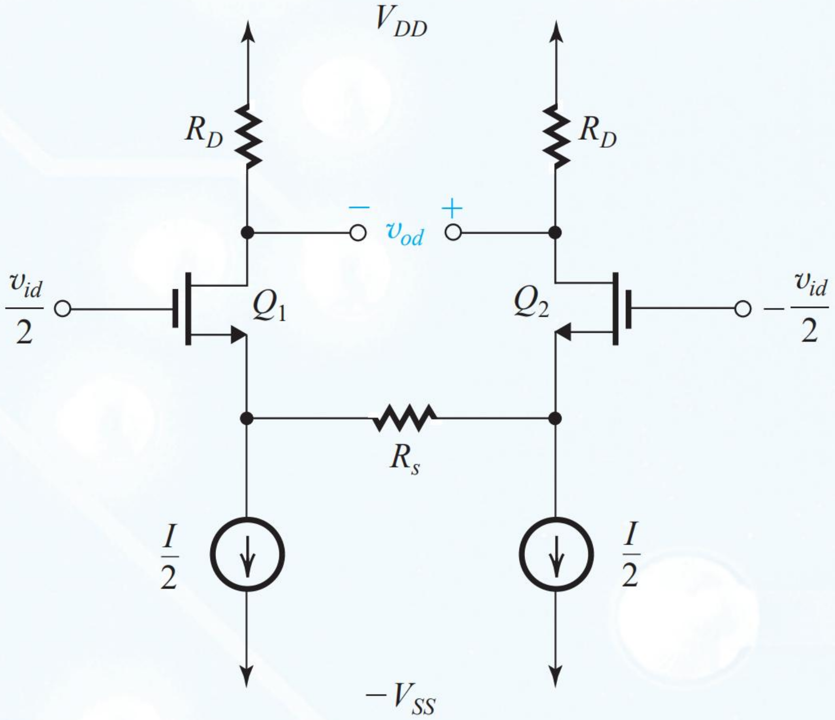
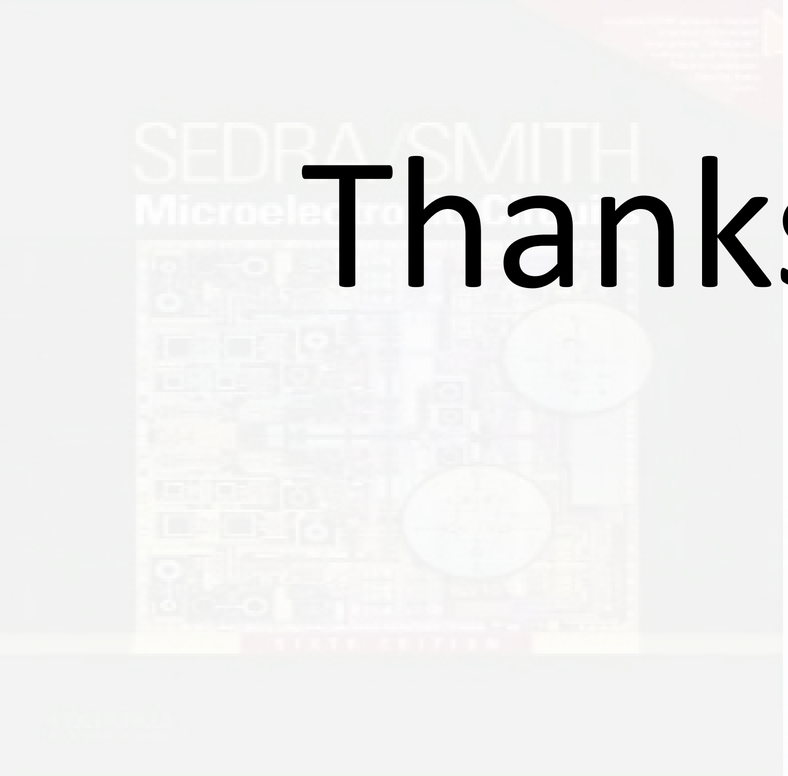


Figure P8.21



Thanks Very Much